



# CROSS-DATASET GENERALISATION OF DEEP LEARNING FOR MALARIA CLASSIFICATION IN MICROSCOPY IMAGES

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LITERATURE REVIEW  
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## 1 PROBLEM STATEMENT

Malaria caused an estimated 282 million cases and 610,000 deaths in 2024 (World Health Organization, [2025](#)). The problem is that early diagnosis still depends on a trained microscopist examining a stained blood slide and identifying the parasite by sight (Rajaraman et al., [2018](#)), and in many remote, malaria-endemic areas, access to good-quality microscopy is limited (World Health Organization, [n.d.](#)).

Several groups depend on that reading: patients waiting on one microscopist who may cover several clinics, the clinician treating them, and the national programme counting cases from the same slides. A missed parasite leaves a patient untreated; a false positive costs unnecessary antimalarials and a missed alternative diagnosis. Sensitivity and specificity are therefore reported separately throughout.

This limited access has driven research into AI systems that can interpret microscopy images automatically. Several such systems now exist, with within-dataset accuracies often above 95% (Rajaraman et al., [2018](#), [2019](#)). Most are tested on one benchmark, the NIH cell images, with cells from the same slide in both training and test sets (Hou et al., [2026](#)). When imaging conditions change, performance drops (Guillon et al., [2025](#); Sukumarran et al., [2024](#)). This project trains CNNs on the NIH cells and tests them unchanged on BBBCo41 and MP-IDB.

## 2 LITERATURE REVIEW

2.1 *State of the Art*

Deep learning for malaria microscopy addresses two tasks. Classification labels a single red blood cell already cut from the smear, while detection locates cells or parasites across the whole image (Rajaraman et al., 2018; Sukumarran et al., 2024). Most NIH studies are classifiers and report per-cell accuracy above 95% (Rajaraman et al., 2018, 2019), although patient-grouped splits lower it (Hou et al., 2026; Rajaraman et al., 2018). Detectors such as YOLOv4 (Sukumarran et al., 2024) and YOLOv5s (Guillon et al., 2025) report mAP or cell-level sensitivity, and their external drops combine localisation and classification errors, so they cannot be compared with a classifier's accuracy or AUC. This study is a classifier, so detector results are used here only for the direction of their drops.

On NIH, lightweight networks match heavy ones. MobileNetV2 performs close to ResNet-50 and VGG-16 (Hemachandran et al., 2023; Loddo et al., 2022), and four architectures from 4.0M to 85.8M parameters, including ViT-B/16, lie within about half a point (Kanamugire & Djoumessi, 2026). On a new site, a pruned YOLOv4 kept more mAP than the full model (90.70% against 81.43%; Sukumarran et al., 2024), but Zhao et al. (2020) tested only VGG16 externally and Hou et al. (2026) only compact models, so NIH-trained light and heavy classifiers have not been compared under the same shift.

External drops also depend on the metric. On BBBCo41, an NIH-trained VGG16's AUC fell from 0.994 to 0.945 (Zhao et al., 2020), while a compact model kept 96.4% sensitivity but reached 18.0% specificity on a test that also changed species, against 82.7% on same-species MP-IDB cells (Hou et al., 2026). With only 2.7–5.1% of BBBCo41 cells parasitised, a high AUC can hide an unusable operating point, and Zhao et al. (2020) split cells randomly, so its baseline is optimistic. Under patient grouping, a detector's sensitivity fell from 79.58% to 38.81–62.07% at three hospitals (Guillon et al., 2025). NIH-trained models also averaged only 60.61% when separating rings from later stages on MP-IDB (Loddo et al., 2022). Wherever it is measured at the operating point, the drop is large.

## 2.2 *Gap*

Crop format here means how a single-cell image is framed. NIH cells were segmented along their outline and placed on black, whereas BBBCo41 and MP-IDB crops are rectangles that keep the plasma and parts of neighbouring cells. Loddo et al. (2022) noted this difference, Zhao et al. (2020) suspected segmentation alongside species and cell overlap, and Hou et al. (2026) listed cell framing with stain colour, optics and scale, but none tested these separately. Because a CNN learns from the whole crop, an NIH-trained model has seen parasites only against a black frame and under one laboratory's Giemsa protocol, and the test crops offer neither.

The literature responds in two ways. One changes the model, through fine-tuning or joint training on labelled target images (Guillon et al., 2025; Loddo et al., 2022) or colour augmentation (Tellez et al., 2019); Hou et al. (2026) trained with colour jitter, yet specificity still collapsed. The other changes the input through stain normalisation, standard in histopathology (Macenko et al., 2009), which needs no target labels. For malaria its effect depends strongly on the method (Xu et al., 2025), and crop format was not considered. No reviewed study varies crop format or stain colour alone with the model fixed, so the cause remains open.

## 2.3 *Baseline and Contribution*

This study adopts the patient-grouped baseline of Hou et al. (2026) and adds a test-time stepwise ablation. Three frozen architectures, one lightweight and two heavy, are evaluated at a fixed threshold (Table 2). The background is first removed from the test cells, which are then colour-matched with Reinhard transfer (Reinhard et al., 2001) or histogram matching. Because the models are not retrained, each change can be attributed to its step. No reviewed study reports the share of the drop recovered per step and architecture. A large recovery would suggest that the models encode transferable parasite morphology, so new sites may not need labelled images, which require expert microscopists. A small one would point to uncorrected factors such as optics, species or life stage.

## 2.4 Research Questions

**MRQ.** To what extent do CNNs trained to classify NIH single-cell images as parasitised or uninfected lose sensitivity on BBBCo41 and MP-IDB, and specificity on BBBCo41, relative to a patient-grouped NIH hold-out set, and which dataset differences explain this drop?

**RQ1.** How large is the drop from the patient-grouped NIH hold-out set, in percentage points, for MobileNetV2, ResNet-50 and VGG-16 applied without fine-tuning to BBBCo41 and MP-IDB, and does the lighter architecture lose less than the heavier ones? A difference between them indicates how far each architecture depends on NIH-specific appearance, which RQ2 then tests directly.

**RQ2.** What share of the RQ1 drop is recovered when test cells are matched to NIH at test time, first in crop format (background removed) and then in stain colour (Reinhard or histogram matching)?

**RQ3.** Does sensitivity on the pooled different-species MP-IDB cells ( $n = 140$ ) differ from *P. falciparum* ( $n = 1,297$ ) by more than the resampling interval, and which BBBCo41 life stages account for the largest share of missed parasites? Because the pooled set is small, and drawn from different slides than the falciparum cells, this question can identify a species effect but cannot exclude one.

| Study                           | Train → test                                             | Models                                                   | Split                                                                  | Within-dataset                     | Cross-dataset                                                                                                  | Stain or crop handling                                                                      |
|---------------------------------|----------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Rajaraman et al. (2018)         | NIH cells                                                | Pretrained CNN features (ResNet-50 best)                 | 5-fold, by cell vs by patient                                          | Acc 98.6% (cell), 95.9% (patient)  | Not tested                                                                                                     | None; patient-level loss put down to staining                                               |
| Zhao et al. (2020)              | NIH → BBBCo41                                            | VGG16 (8 CNNs compared)                                  | Random 5-fold cross-validation                                         | AUC 0.994                          | AUC 0.945                                                                                                      | None; suspects segmentation, species, overlap                                               |
| Loddo et al. (2022)             | NIH → MP-IDB <i>P. falciparum</i> crops                  | 11 CNNs incl. MobileNetV2, ResNet-50, VGG-16             | NIH 80/10/10; MP-IDB 80/20 (stage task), 50/50 (cross-dataset), 5-fold | Acc 97.31% (MobileNetV2, binary)   | Stage task: 60.61% mean zero-shot, 92.59% fine-tuned                                                           | One colour balance for all images                                                           |
| Sukumarran et al. (2024)        | MP-IDB → MRC-UNIMAS                                      | YOLOv4 and pruned variants                               | Held-out images                                                        | mAP 93.87%                         | mAP 81.43% (pruned: 90.70%)                                                                                    | None; authors attribute the drop to overfitting and note <i>P. knowlesi</i> in the test set |
| Guillon et al. (2025)           | NIH photographs → 3 hospitals                            | YOLOv5s                                                  | Patient-grouped 5-fold                                                 | Cell sensitivity 79.58%            | Cell sensitivity 38.81–62.07%                                                                                  | Joint training, fine-tuning                                                                 |
| Hou et al. (2026)               | NIH → BBBCo41, MP-IDB                                    | 8 CNNs; external test on compact models only             | Slide-disjoint 70/15/15                                                | Acc 95.61% (97.06% per-cell split) | BBBCo41 sens 96.4%, spec 18.0% (slide-disjoint; spec 37.2% if trained per cell); MP-IDB sens 74.0%, spec 82.7% | Fixed Giemsa stain stream, colour jitter; no normalisation                                  |
| Kanamugire and Djoumessi (2026) | NIH cells, corrupted copies                              | ResNet-18, EfficientNet-B0, MobileNet-v3-Large, ViT-B/16 | About 63/37, stratified 5-fold                                         | Acc 0.958–0.963                    | No external set; noise and blur only                                                                           | None                                                                                        |
| Xu et al. (2025)                | In-house <i>P. falciparum</i> smears → 2 other platforms | ResNet-18 (stage classification)                         | By imaging platform                                                    | Not reported                       | Stage acc 21.32% (none), 29.34% (Reinhard), 38.81% (Vahadane), 48.66% (Beer-LaNet)                             | Compares stain normalisation methods                                                        |

Table 1: In-dataset and cross-dataset results for the malaria studies discussed, and the stain normalisation comparison closest to RQ2.

| Term                      | Definition                                                                                                                                                                                          |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Task                      | Binary classification of single-cell images; parasitised is the positive class                                                                                                                      |
| Decision rule             | A cell is classified as parasitised when the model's probability is at least 0.5. The threshold is fixed before testing, and no test set is used for fine-tuning                                    |
| Sensitivity (recall)      | Parasitised cells classified as parasitised, divided by all parasitised cells                                                                                                                       |
| Specificity               | Uninfected cells classified as uninfected, divided by all uninfected cells. Not measurable on MP-IDB, whose annotations mark parasites only; Hou et al. (2026) segmented their own uninfected cells |
| AUC                       | Threshold-free check on the NIH hold-out set and on BBBCo41                                                                                                                                         |
| Drop                      | Value on the NIH hold-out set minus value on a test set, in percentage points, per model and test set                                                                                               |
| Share recovered (RQ2)     | Value after matching minus value before matching, divided by the drop                                                                                                                               |
| Uncertainty               | 95% intervals from resampling source images, because cells from one image are not independent                                                                                                       |
| NIH hold-out set          | Patients not used for training: 2,055 parasitised and 2,122 uninfected cells                                                                                                                        |
| BBBCo41 site_a and site_b | site_a: 2,149 parasitised and 77,420 uninfected cells; site_b: 303 and 5,614. Ring, trophozoite, schizont and gametocyte cells count as parasitised; difficult and leukocyte cells are excluded     |
| MP-IDB                    | <i>P. falciparum</i> , the same species as NIH: 1,297 parasitised cells. <i>P. malariae</i> , <i>P. ovale</i> and <i>P. vivax</i> pooled: 140 parasitised cells                                     |

Table 2: Measures and test sets used in the research questions.

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